

MODULE-2:

Here is a summary of the key topics in the provided document.

Module 2: Cloud Economics and Billing Overview

This module covers the following topics:

- Fundamentals of pricing
- Total Cost of Ownership (TCO)
- AWS Organizations
- AWS Billing and Cost Management
- Technical Support

Section 1: Fundamentals of Pricing

AWS Pricing Philosophy

- AWS pricing is based on a "pay-as-you-go" model, meaning users only pay for the services they use, without upfront expenses or long-term contracts.
- AWS aims to pass savings from its economies of scale to users by lowering prices.
- Since 2006, AWS has lowered prices 75 times (as of September 2019).
- Users can gain hands-on experience with the AWS platform through the AWS Free Tier, which is free for new customers for up to one year.

AWS Pricing Models

Three fundamental drivers of cost with

AWS Compute

- Charged per hour/second
- Varies by instance type

*Linux only

Storage

- Charged typically per GB

Data transfer

- Outbound is aggregated and charged
- Inbound has no charge (with some exceptions)

- Charged typically per GB
- **Pay for what you use:** Users only pay for consumed services.
- **Pay less when you reserve:** Users can save up to 75% by investing in Reserved Instances (RIs) for services like Amazon EC2 and Amazon RDS.
 - RI options include All Upfront (AURI), Partial Upfront (PURI), and No Upfront Payments (NURI).
- **Pay less when you use more:** Volume-based discounts are available, with tiered pricing for services like Amazon S3, Amazon EBS, and Amazon EFS.
- **Pay even less as AWS grows:** AWS passes on savings from its growth and efficiency improvements to users through lower pricing.
- **Custom pricing:** Available for high-volume projects with unique requirements.

Services with No Charge

AWS offers several services for no additional charge, including:

- Amazon VPC
- AWS Identity and Access Management (IAM)
- Consolidated Billing
- AWS Elastic Beanstalk
- AWS CloudFormation
- Automatic Scaling
- AWS OpsWorks

Important Considerations

- While some services are free, charges may apply for other AWS services used with them.
- Inbound data transfer and data transfer between services within the same AWS Region are typically free, but exceptions exist.
- Outbound data transfer is charged based on aggregated usage across services.

Section 2: Total Cost of Ownership (TCO)

On-Premises vs. Cloud

- On-premises infrastructure involves fixed costs for facilities, hardware, licenses, and maintenance staff, while scaling can be expensive and time-consuming.
- Cloud infrastructure, purchased from a service provider, allows for a pay-as-you-go model, making scaling simpler and costs easier to estimate.

Understanding TCO

- TCO is a financial estimate encompassing the direct and indirect costs of a system, used to compare running an infrastructure or workload on-premises versus on AWS.
- It helps with budgeting and building a business case for cloud migration.

TCO Considerations

TCO calculations for on-premises solutions should include all direct and indirect costs:

- Direct costs: Server hardware and software, facilities (space, power, cooling)
- Indirect costs: Storage hardware, administration, network infrastructure
- IT labor costs: Server administration, storage administration, network administration

AWS Pricing Calculator

- The AWS Pricing Calculator is a tool for estimating monthly AWS costs and identifying cost reduction opportunities.
- It allows users to model solutions before building them and explore price points and calculations.
- Users can find suitable instance types and contract terms and create groups of services for better organization.

Additional Benefit Considerations

- **Hard benefits:** Reduced spending on compute, storage, networking, security; reductions in hardware and software purchases; reductions in operational costs; reduction in operations personnel
- **Soft benefits:** Reuse of services and applications; increased developer productivity; improved customer satisfaction; agile business processes; increased global reach

Case Study: Delaware North

- Delaware North, a global food and hospitality company, moved its on-premises data center to AWS to meet the demand for rapid solution deployment and to avoid constant equipment upgrades.
- They achieved a cost saving of \$3.5 million over five years, reducing their server hardware by 90%.
- In addition to cost savings, they experienced benefits such as enhanced security compliance, improved disaster recovery, and faster deployment times for new services.

Section 3: AWS Organizations

Introduction

- AWS Organizations is a free account management service that enables users to consolidate multiple AWS accounts into a central organization for management.

Benefits

- Centrally managed access policies across multiple AWS accounts
- Controlled access to AWS services

- Automated AWS account creation and management
- Consolidated billing across multiple AWS accounts

Terminology

- **Root:** The main container for all accounts in an organization.
- **Organizational Unit (OU):** A container for accounts within a root, allowing hierarchical organization.
- **Policy:** A set of rules applied to accounts or OUs to control access and permissions.

Key Features and Benefits

- **Policy-based account management:** Service Control Policies (SCPs) allow centralized control of AWS services across accounts.
- **Group-based account management:** Create groups of accounts and apply policies to ensure consistency.
- **API-driven automation:** Simplify account management with APIs for account creation and management.
- **Consolidated billing:** Streamline billing with a single payment method and combined view of charges.

Security

- AWS Organizations complements IAM policies by using SCPs to control service access for accounts or groups of accounts, including the root user.

Setup

1. Create an organization with your current AWS account as the primary account.
2. Create organizational units (OUs) and place member accounts in those OUs.
3. Create service control policies (SCPs) to define restrictions on actions for member accounts.
4. Test the organization's policies to ensure they function as intended.

Limits

- Restrictions on names for accounts, OUs, roots, and policies (Unicode characters, maximum 250 characters)
- Specific limits on the number of accounts, OUs, policies, and other entities.

Accessing AWS Organizations

- **AWS Management Console:** Browser-based interface for managing organizations and resources.
- **AWS Command Line Interface (AWS CLI):** Issue commands for AWS Organizations tasks.

- Software Development Kits (SDKs): Libraries and sample code for various programming languages.
- HTTPS Query API: Programmatic access to AWS Organizations using HTTPS requests.

Section 4: AWS Billing and Cost Management

Introduction

- AWS Billing and Cost Management is a service used for:
 - Paying AWS bills
 - Monitoring usage
 - Budgeting costs
 - Forecasting future costs and usage

AWS Billing Dashboard

- Provides an overview of month-to-date AWS expenditure.
- Identifies services with the highest expenditure.
- Shows cost trends.
- Includes graphs like "Spend Summary" and "Month-to-Date Spend by Service."

Tools

- **AWS Bills:** Detailed breakdown of AWS costs and usage, including the monthly bill.
- **AWS Cost Explorer:** Visualizes, understands, and manages AWS costs and usage over time.
 - Provides a default report visualizing costs and usage for top cost-incurring services.
 - Offers forecasting for the coming month.
 - Allows viewing cost data for the past 13 months.
 - Helps identify cost problem areas.
 - Shows the most used services and metrics like Availability Zone traffic and linked account usage.
- **AWS Budgets:** Allows users to set budgets, track costs, and receive notifications when budgets are exceeded.
 - Uses Cost Explorer for cost visualization.
 - Provides forecasts of estimated costs.
 - Can be tracked at monthly, quarterly, or yearly levels.
 - Sends budget alerts via email or Amazon SNS.
- **AWS Cost and Usage Report:** Comprehensive information about AWS costs and usage in a single location.
 - Lists usage for each service category, including hourly or daily line items.
 - Provides information on activated taxes.
 - Allows publishing billing reports to an S3 bucket, updated once a day.

Section 5: Technical Support

AWS Support

- Offers a combination of tools and expertise to support customers at different stages of AWS adoption.
- Provides assistance with:
 - Experimenting with AWS
 - Production use of AWS
 - Business-critical use of AWS

Proactive Guidance

- **Technical Account Manager (TAM):** Serves as the primary point of contact for proactive guidance, architectural reviews, and ongoing communication.
- **AWS Trusted Advisor:** Online resource that checks for opportunities to reduce costs and increase productivity by following best practices.

Account Assistance

- **AWS Support Concierge:** A billing and account expert who provides analysis on billing and account-level inquiries.

Support Plans

- **Basic Support:** Resource Center access, Service Health Dashboard, FAQs, discussion forums, support for health checks.
- **Developer Support:** Support for early development on AWS.
- **Business Support:** Support for customers running production workloads on AWS.
- **Enterprise Support:** Support for customers running business and mission-critical workloads. Includes a Technical Account Manager (TAM).

Case Severity and Response Times

- Case severity levels: Critical, Urgent, High, Normal, Low
- Response times vary depending on the support plan and case severity.

Module Wrap-up

Module Summary

The module covered:

- Fundamentals of AWS pricing
- TCO concepts
- AWS Pricing Calculator estimate
- Billing dashboard
- Technical support options and costs

